Specifications

- MOSFET: AO3400A N-Channel Enhancement MOSFET
- Configuration: Low-side switching
- Operating Voltage: 2–12 VDC
- Recommended Continuous Load: Up to 2 A
- Gate Drive: Compatible with 3.3 V and 5 V logic
- Indicator: Optional output voltage LED
- Connector: 4-pin header
    - DR – Drain
    - GT – Gate
    - SR – Source
    - GND – LED ground (optional)

Note: The GND pin is only required for the indicator LED.
The MOSFET itself operates normally without it.

Features

- Logic-level N-channel MOSFET module
- Clearly labeled Drain, Gate, and Source terminals
- Silkscreen MOSFET symbol showing the intrinsic body diode
- Optional output voltage indicator LED
- LED can be disabled simply by leaving the GND pin unconnected
- Compatible with 3.3 V and 5 V logic outputs
- Suitable for breadboards and educational experiments
- Compact module for learning MOSFET operation and switching circuits

Applications

- LED control
- Relays
- Solenoids
- Small DC motors
- Low-side switching
- Logic-controlled loads
- MOSFET learning and prototyping

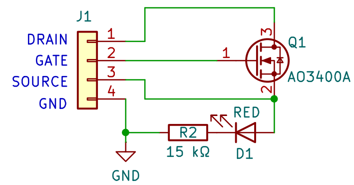
Notes

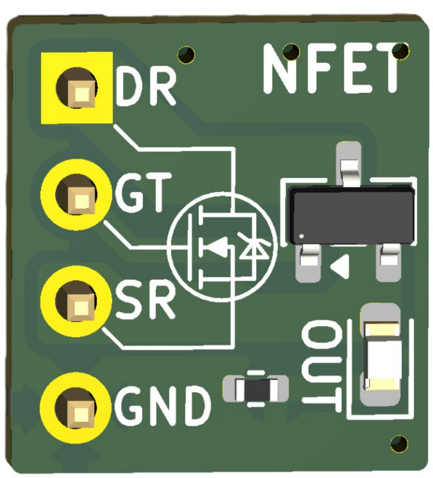
- Operating voltage (2–12 V) refers to the load circuit voltage, not the gate drive.
- The 2 A rating is a recommended module limit, accounting for PCB size and thermal considerations rather than the MOSFET's absolute maximum rating.
- The LED indicates the monitored output node is energized; it does not indicate that the MOSFET is conducting.

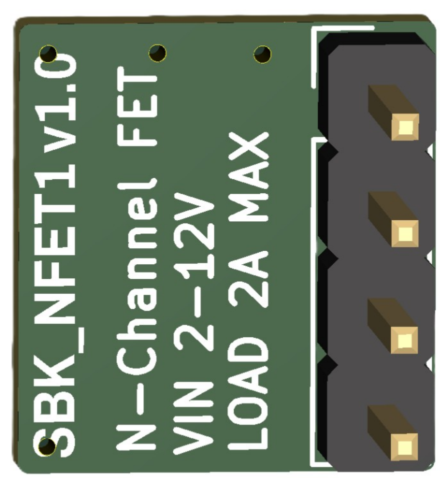
Important
- An external flyback diode is required when driving inductive loads (relays, solenoids, motors).
- Connect the diode across the load, not across the MOSFET.
- Resistive loads (LEDs, heaters, resistors) do not require a flyback diode.

BOM

Reference	Qty	Value	Package
D1	1	RED	0603
J1	1	1 x 4 pin header	2.54 mm
Q1	1	AO3400A	SOT-23
R2	1	15 kΩ	0402







SBK_NFET1 – N-Channel FET Module

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